

Realization of Quantum Unitary Gates using Born Scattering

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Abstract—We develop the Born series (Lippmann–Schwinger) formalism for potential scattering and the associated Møller wave operators and scattering matrix $S(E)$, with the aim of investigating the fixed-energy scattering matrix as a candidate realization of a unitary quantum gate acting on plane-wave momentum states. The present manuscript develops the underlying scattering-theoretic framework and establishes that $S(E)$ is unitary on the space of fixed-energy states $L^2(S^2)$. The novelty of this work is establishing a theoretical framework that interprets the energy-shell scattering matrix as a finite-dimensional quantum gate by restricting it to a discrete computational basis. A procedure for constructing an explicit finite-dimensional gate matrix from $S(E)$ is outlined in the concluding section, together with the corresponding inverse potential-design problem.

I. INTRODUCTION

Quantum gate synthesis is conventionally addressed either by engineering a time-dependent Hamiltonian whose unitary propagator implements a target gate [2]—a controllability problem that can be characterised group-theoretically through root-space decompositions of the Lie algebra $su(N)$ [26]—or by decomposing a target unitary into a sequence of elementary gates realised on a fixed physical platform (superconducting qubits, trapped ions, linear optics, etc.) [25]. Both routes treat the interaction as something that is switched on for a controlled duration. The first approach requires active control of the interaction strength and timing, often requiring rapid pulse sequences or carefully tuned electromagnetic fields; a concrete instance is the synthesis of a target gate by a small time-dependent perturbation of a free particle confined in a box under an applied electric field [27]. The second approach decomposes a target gate into a sequence of elementary operations, each corresponding to a native gate available on the physical platform, which may result in deep quantum circuits and accumulation of gate errors; algorithmic primitives that place a premium on precise gate synthesis in this setting include the generalized quantum Fourier transforms used in quantum phase estimation [3]. Both methods implicitly assume that quantum information is encoded in internal degrees of freedom—spin states, electronic states, or photon polarization—and that the implementation is fundamentally dynamic: the quantum system must be prepared, interacted for a controlled time, and then measured or further processed.

Potential scattering suggests a third route that is fundamentally different in character. Consider a particle prepared in a definite-momentum (plane-wave) state $|\psi_i\rangle$ and sent through a static potential $V(r)$ that is fixed in space and time. As the particle evolves according to the Schrödinger equation, it interacts with the potential and is scattered. At large times, the scattered state emerges in an outgoing plane-wave state $|\psi_0\rangle$ related to the incident state via

$$|\psi_0\rangle = S(E)|\psi_i\rangle$$

, where $S(E)$ is the energy-shell scattering matrix at the fixed collision energy $E = \hbar^2 k^2 / 2m$ of the incident particle. Because $S(E)$ is unitary on the space of fixed-energy states [4], it is natural to ask whether a suitably designed potential $V(r)$ can make $S(E)$, restricted to a finite-dimensional subspace spanned by a discrete set of incident directions $\{\hat{n}_1, \dots, \hat{n}_N\}$, coincide with a target N -dimensional unitary gate. In this picture the “gate” is not switched on and off in time; it is encoded once, permanently, in the geometric and functional form of the scattering potential $V(r)$. Different logical basis states are represented by different incident directions (or, equivalently, different points on the momentum sphere S^2) at a fixed energy $E = \hbar^2 k^2 / 2m$. The quantum gate operation is then simply the scattering process itself: a particle entering in state $|\psi_{i,\ell}\rangle$ corresponding to incident direction $\hat{n}_\ell$ emerges as a superposition $\sum_{\ell'} U_{\ell\ell'} |\psi_{0,\ell'}\rangle$, where the matrix $U_{\ell\ell'} = \langle \psi_{0,\ell'} | S(E) | \psi_{i,\ell} \rangle$ is the implemented unitary gate. Fig. 1 illustrates this scattering-based gate-encoding picture.

This scattering-based approach to quantum information has several conceptual and practical advantages. First, the gate is encoded in a static spatial structure rather than a time-dependent control field, which may be easier to fabricate and maintain in certain physical systems. Second, different logical basis states are encoded in different spatial momentum directions rather than in distinct internal quantum numbers, which may reduce coupling to decoherence channels associated with internal state mixing; the discrete set $\{\hat{n}_1, \dots, \hat{n}_N\}$ of incident directions used here is only one choice of overcomplete basis for this purpose, and generalized coherent states [5], [11], including those built from oscillator-type ladder-operator realizations [12], offer an alternative encoding that may be explored in future extensions of this framework. Third, the theoretical framework is grounded in the well-developed mathematical machinery of quantum scattering theory [17], [22] and of general operator-theoretic and functional-analytic methods [1], providing rigorous tools for analysis, design, and optimization, and resting on the perturbation theory of linear

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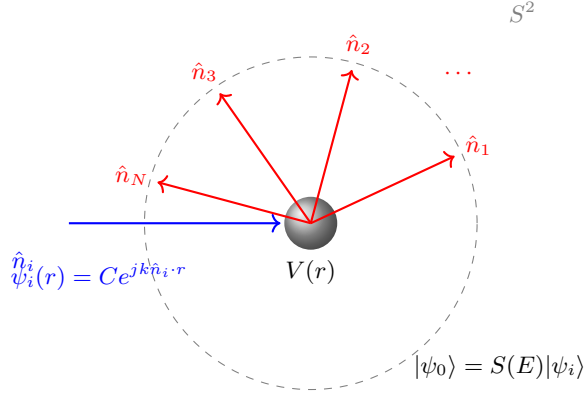


Fig. 1. Schematic of the scattering-based gate encoding. A plane wave of fixed energy $E = \hbar^2 k^2 / 2m$, incident along $\hat{n}_i$, scatters off the static potential $V(r)$ fixed at the origin; the outgoing state is a superposition over a discrete set of directions $\{\hat{n}_1, \dots, \hat{n}_N\}$ on the momentum sphere S^2 , with amplitudes $U_{\ell\ell'} = \langle \psi_{i\ell'} | S(E) | \psi_{i\ell} \rangle$ forming the realized N -dimensional quantum gate. Each direction $\hat{n}_\ell$ encodes one logical basis state; unlike a time-dependent-control gate, the gate here is fixed once and for all by the static geometry of $V(r)$.

operators [10] and the foundational bra-ket operator formalism of nonrelativistic quantum mechanics [6], [9].

The scattering-based approach to quantum information is closely related in spirit to recent developments in quantum computation via scattering channels [7], [18], in which qubits are encoded in the channels of a scattering process rather than in internal degrees of freedom; this remains an active line of inquiry, as illustrated by a recent demonstration that every two-particle $SU(N)$ -invariant scattering amplitude can be assembled from a minimal set of three unitary quantum gates [32]. Classical scattering theory provides a rigorous mathematical foundation through the formalism of the Lippmann–Schwinger equation [13], which relates incoming and outgoing asymptotic states via the resolvent form of the Schrödinger equation, and whose scope continues to be extended, for instance to two-dimensional multiple-slit diffraction geometries [33] and to magnetotransport modelling in double quantum wires [31]. This equation expresses the scattered wave function as a superposition of the incident plane wave plus a scattering contribution determined by the potential. The Møller wave operators [14], originally introduced to compare the dynamics of interacting systems with free-particle evolution, serve as the fundamental building blocks for constructing the scattering matrix. They encode the relationship between asymptotic free states at $t \rightarrow -\infty$ (incoming states) and the actual scattering states at finite times, and similarly for outgoing states at $t \rightarrow +\infty$. More recently, alternative formulations using the Dyson series expansion [20] have provided computational advantages for systems with time-varying potentials and have been applied to quantum electrodynamics-based gate design [19].

The Born series expansion [15], [16] provides a systematic perturbative approach to solving the scattering problem. In this expansion, the scattered wave function is written as a power series in the coupling strength of the potential, and each term represents successive scattering events. The first-order Born

approximation retains only the leading-order scattering term, corresponding to a single interaction of the incoming particle with the potential. Higher-order terms account for multiple scatterings, in which the particle scatters off the potential, propagates, and scatters again. The mathematical framework relies on the limiting absorption principle and Green’s function methods for solving the inhomogeneous Helmholtz equation [8], [21], which describe the outgoing wave boundary condition essential for scattering theory.

The scattering matrix $S(E) = \Omega_+(E)^* \Omega_-(E)$, constructed from the Møller operators, encodes all scattering information at a fixed energy E . Its matrix elements $\langle \psi_{i2} | S(E) | \psi_{i1} \rangle$ describe the amplitude for a particle to scatter from an incident direction $\hat{n}_1$ to an outgoing direction $\hat{n}_2$. For the scattering matrix to serve as a quantum gate, these matrix elements must exhibit several properties: (1) unitarity, to ensure probability conservation; (2) spatial localization, so that the gate acts primarily on a restricted set of incident directions; (3) controllability, so that by varying the potential $V(r)$ one can engineer different gate matrices; and (4) discretizability, so that the continuum of incident directions can be reduced to a finite computational basis without significant loss of fidelity.

The present paper carries out the necessary scattering-theory groundwork required to establish this framework rigorously. Section II derives the Born series for the scattered state ψ_f order by order in the coupling strength ϵ , resums it into the exact resolvent form $|\psi_f\rangle = [I - \epsilon(H_1 - E)^{-1}V]|\psi_i\rangle$, sets up the Møller wave operators $\Omega_\pm(E)$, and obtains the scattering matrix $S(E) = \Omega_+(E)^* \Omega_-(E)$ along with its plane-wave matrix elements $\langle \psi_{i2} | S(E) | \psi_{i1} \rangle$. These matrix elements are derived both in the far-field (radiation-zone) limit, which simplifies the asymptotic form, and without that approximation, to retain accuracy for near-field configurations. The key result is the explicit form of $S(E)$ to first order in ϵ , which reveals the structure of the scattering amplitude and provides the foundation for gate design. Section III shows that this static, energy-shell construction is in fact a direct generalization of the time-dependent Dyson-series gate-design method of Gautam *et al.* [34], in which a target unitary is realized by weakly driving a quantum harmonic oscillator with a time-varying field: we exhibit the term-by-term correspondence between the two formalisms, extend the Lagrange-multiplier variational construction of [34] from the time-dependent field $E(t)$ to the static potential $V(r)$, and give a first numerical illustration of the resulting gate, for a spherically symmetric potential and a two-direction (single-qubit) encoding, comparing its noise-to-signal-ratio behavior directly against that of the oscillator-based route of [34]. Section IV (Conclusions) discusses how the matrix elements $\langle \psi_{i\ell} | S(E) | \psi_{i\ell'} \rangle$ for a general discrete set of incident directions $\{\hat{n}_1, \dots, \hat{n}_N\}$ can be assembled into an explicit finite-dimensional gate matrix $U_{\ell\ell'}$, and identifies the open questions that remain to be addressed.

II. PROBLEM FORMULATION

Consider an incident projectile plane-wave state

$$\psi_i(r) = C \exp(j\hat{k}\hat{n}_i \cdot r), \quad (1)$$

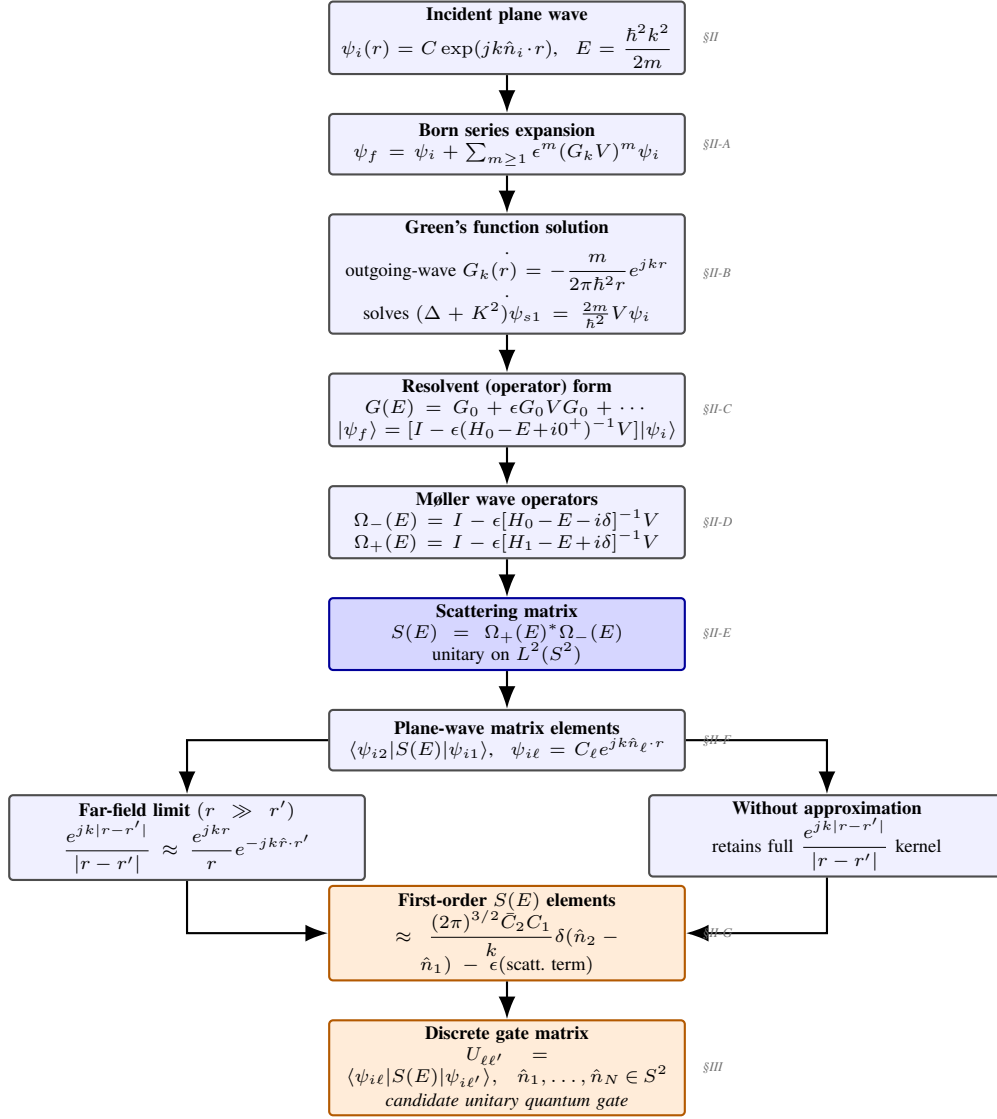


Fig. 2. Block diagram of the mathematical pipeline used to formulate the scattering matrix from Born scattering theory, and of how it is proposed to be turned into a finite-dimensional quantum gate. Starting from an incident plane-wave state, the Born series (Sec. II-A) is resummed into an exact resolvent (operator) form (Sec. II-C), which defines the Møller wave operators $\Omega_{\pm}(E)$ (Sec. II-D) and hence the energy-shell scattering matrix $S(E) = \Omega_+(E)^* \Omega_-(E)$ (Sec. II-E), unitary on $L^2(S^2)$. Its plane-wave matrix elements (Sec. II-F) are evaluated both in the far-field limit and without approximation (Sec. II-G), giving the explicit first-order $S(E)$; restricting this to a discrete set of incident directions yields the finite-dimensional gate matrix $U_{\ell\ell'}$ discussed in Sec. III.

with projectile energy

$$E = \frac{\hbar^2 k^2}{2m}, \quad (2)$$

where $\hat{n}_i$ is the incident direction. This satisfies the free Schrödinger equation

$$-\frac{\hbar^2}{2m} \Delta \psi_i = E \psi_i. \quad (3)$$

A. Born Series

Let the interaction potential energy be $\epsilon V(r)$ and write the final (scattered) state as a formal power series in ϵ :

$$\psi_f(r) = \psi_i(r) + \epsilon \psi_{s1}(r) + \epsilon^2 \psi_{s2}(r) + O(\epsilon^3). \quad (4)$$

Substituting into

$$\left[-\frac{\hbar^2}{2m} \Delta + \epsilon V \right] \psi_f = E \psi_f \quad (5)$$

and collecting powers of ϵ gives, at order ϵ^0 , the free equation for ψ_i above (automatically satisfied), and at orders ϵ^1, ϵ^2 ,

$$-\frac{\hbar^2}{2m} \Delta \psi_{s1} + V \psi_i = E \psi_{s1}, \quad (6)$$

$$-\frac{\hbar^2}{2m} \Delta \psi_{s2} + V \psi_{s1} = E \psi_{s2}. \quad (7)$$

Writing $K = k$ and rearranging as inhomogeneous Helmholtz equations,

$$(\Delta + K^2) \psi_{s1} = \frac{2m}{\hbar^2} V \psi_i, \quad (8)$$

$$(\Delta + K^2) \psi_{s2} = \frac{2m}{\hbar^2} V \psi_{s1}. \quad (9)$$

B. Green's Function Solution

Using the outgoing-wave Green's function of the Helmholtz operator, the first-order scattered wave is

$$\psi_{s1}(r) = -\frac{m}{2\pi\hbar^2} \int \frac{\exp(jk|r-r'|)}{|r-r'|} \times V(r') \psi_i(r') d^3r' = \int G_k(r-r') V(r') \psi_i(r') d^3r', \quad (10)$$

with

$$G_k(r) = -\frac{m}{2\pi\hbar^2 r} \exp(jkr), \quad (11)$$

and, iterating,

$$\psi_{s2}(r) = \iint G_k(r-r') V(r') G_k(r'-r'') \times V(r'') \psi_i(r'') d^3r' d^3r''. \quad (12)$$

Formally, $\psi_{sm} = (G_k V)^m \psi_i$ for $m = 1, 2, 3, \dots$, so the exact solution is the Born series

$$\psi_f = \psi_i + \sum_{m=1}^{\infty} \epsilon^m (G_k V)^m \psi_i. \quad (13)$$

For $\psi \in L^2(\mathbb{R}^3)$, boundedness of $G_k V$ follows from

$$\|G_k V \psi\|^2 \leq \left[\int |G_k(r-r') V(r')|^2 d^3r d^3r' \right] \times \left[\int |\psi(r')|^2 d^3r' \right], \quad (14)$$

which is finite provided $\int |V(r')|^2 d^3r' < \infty$ and $\int |G_k(r)|^2 d^3r < \infty$; the latter integral in fact diverges for the free Green's function ($\int r^{-2} \cdot 4\pi r^2 dr = \infty$), but $\|G_k V\|$ can still be made finite for suitable V (e.g. compactly supported or sufficiently rapidly decaying potentials).

C. Resolvent (Operator) Form

Writing $H_0 = -\frac{\hbar^2}{2m} \Delta$, one finds

$$G_k(r) = -\frac{m}{2\pi\hbar^2} \left\langle r \left| \frac{1}{E + i0^+ - H_0} \right| r' \right\rangle. \quad (15)$$

More generally, the full Green's function of the interacting Hamiltonian $H = H_0 + \epsilon V$ is

$$G(E) = [E - H + i0^+]^{-1}. \quad (16)$$

By the resolvent identity,

$$G(E) = G_0(E) + G_0(E) \epsilon V G(E), \quad (17)$$

which is equivalent to the Born series when iterated:

$$G(E) = G_0(E) + \epsilon G_0(E) V G_0(E) + \epsilon^2 G_0(E) V G_0(E) V G_0(E) + \dots \quad (18)$$

The scattered state is related to this resolvent form by

$$|\psi_f\rangle = [I - \epsilon(H_0 - E + i0^+)^{-1} V] |\psi_i\rangle. \quad (19)$$

D. Møller Wave Operators

The asymptotic behavior of a scattering state at large negative time is

$$\lim_{t \rightarrow -\infty} e^{iH_0 t} e^{-iHt} |\psi_f\rangle = |\psi_i\rangle \quad (20)$$

(more precisely, in the weak sense). This defines the incoming Møller wave operator,

$$\Omega_-(E) = \lim_{t \rightarrow -\infty} e^{iH_0 t} e^{-iHt} \Big|_{\text{on states of energy } E}. \quad (21)$$

By differentiation with respect to time and using the Schrödinger equation,

$$|\psi_f\rangle = \Omega_- |\psi_i\rangle, \quad \Omega_- = I - \epsilon[H_0 - E - i\delta]^{-1} V, \quad (22)$$

where the notation $[H_0 - E \pm i\delta]^{-1}$ is shorthand for $(E - H_0 \mp i0^+)^{-1}$. Similarly, the outgoing wave operator (defined by the limit as $t \rightarrow +\infty$) is

$$\Omega_+(E) = I - \epsilon[H_1 - E + i\delta]^{-1} V, \quad (23)$$

where we use the notation $H_1 \equiv H_0$ for notational parallelism. The two are related by the adjoint: $\Omega_+^* = \lim_{t \rightarrow +\infty} e^{iHt} e^{-iH_0 t}$, and

$$\Omega_+(E)^* \Omega_+(E) = I \quad (24)$$

(isometry). Thus

$$|\psi_0\rangle = \Omega_+(E)^* |\psi_f\rangle = \Omega_+(E)^* \Omega_-(E) |\psi_i\rangle. \quad (25)$$

The Born approximation of Sec. II-B is the finite-order truncation of $|\psi_f\rangle = \Omega_-(E) |\psi_i\rangle$.

E. Scattering Matrix

Since $\Omega_+(E)$ is an isometry, $\Omega_+(E)^* \Omega_+(E) = I$, so $|\psi_0\rangle = \Omega_+(E)^* |\psi_f\rangle = \Omega_+(E)^* \Omega_-(E) |\psi_i\rangle$, which defines the scattering matrix at energy E ,

$$S(E) = \Omega_+(E)^* \Omega_-(E), \quad |\psi_0\rangle = S(E) |\psi_i\rangle. \quad (26)$$

$S(E)$ is unitary on the space of fixed-energy states, i.e. states defined on the unit sphere, $L^2(S^2)$. To leading order in ϵ ,

$$\begin{aligned} S(E) &= [I - \epsilon[H_1 - E + i\delta]^{-1} V]^* \\ &\quad \times [I - \epsilon[H_1 - E - i\delta]^{-1} V] \\ &= I - \epsilon \left\{ V[H_0 - E - i\delta]^{-1} \right. \\ &\quad \left. + [H_0 - E - i\delta]^{-1} V \right\} + O(\epsilon^2). \end{aligned} \quad (27)$$

F. Plane-Wave Matrix Elements

Take two incident plane waves

$$\psi_{i1}(r) = C_1 \exp(jk\hat{n}_1 \cdot r), \quad (28)$$

$$\psi_{i2}(r) = C_2 \exp(jk\hat{n}_2 \cdot r). \quad (29)$$

Their Fourier transforms are

$$\hat{\psi}_{i1}(k') = (2\pi)^{3/2} C_1 \delta(k' - k\hat{n}_1), \quad (30)$$

$$\hat{\psi}_{i2}(k') = (2\pi)^{3/2} C_2 \delta(k' - k\hat{n}_2), \quad (31)$$

so that

$$\begin{aligned}\langle\psi_{i2}|\psi_{i1}\rangle &= \int \bar{\psi}_{i2}(k')\psi_{i1}(k') d^3k' \\ &= (2\pi)^3 \bar{C}_2 C_1 \delta(k(\hat{n}_2 - \hat{n}_1)) \\ &= \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1),\end{aligned}\quad (32)$$

and

$$\begin{aligned}\langle\psi_{i2}|V|\psi_{i1}\rangle &= \bar{C}_2 C_1 \int V(r) \\ &\quad \times \exp(-jk(\hat{n}_2 - \hat{n}_1) \cdot r) d^3r \\ &= (2\pi)^{3/2} \bar{C}_2 C_1 \hat{V}(k(\hat{n}_2 - \hat{n}_1)).\end{aligned}\quad (33)$$

By the Riemann–Lebesgue lemma, $\lim_{k \rightarrow \infty} \hat{V}(k(\hat{n}_2 - \hat{n}_1)) = 0$, so this term is negligible at high energy. The Born term is

$$\begin{aligned}\langle\psi_{i2}|[H_0 - E]^{-1}|\psi_{i1}\rangle &= -\frac{2m}{\hbar^2} \langle\psi_{i2}|[\Delta + k^2]^{-1}V|\psi_{i1}\rangle \\ &= \frac{m}{2\pi\hbar^2} \int \frac{\exp[jk|r - r'|]}{|r - r'|} \\ &\quad \times \bar{\psi}_{i2}(r)V(r')\psi_{i1}(r') d^3r d^3r'.\end{aligned}\quad (34)$$

Since ψ_{i1}, ψ_{i2} are concentrated at $r \rightarrow \infty$ while $V(r)$ is concentrated near the origin, $\langle\psi_{i2}|V|\psi_{i1}\rangle \approx 0$, and

$$\begin{aligned}\langle\psi_{i2}|S(E)|\psi_{i1}\rangle &\approx \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1) \\ &\quad - \frac{m\epsilon}{2\pi\hbar^2} \bar{C}_2 C_1 \int \frac{\exp(jk|r - r'|)}{|r - r'|} \\ &\quad \times \exp(jk[\hat{n}_1 \cdot r' - \hat{n}_2 \cdot r]) V(r') d^3r' d^3r.\end{aligned}\quad (35)$$

In the far field, $r \gg r'$, so

$$\frac{\exp(jk|r - r'|)}{|r - r'|} \approx \frac{e^{jkr}}{r} \exp(-jk\hat{r} \cdot r'),\quad (36)$$

giving

$$\begin{aligned}\langle\psi_{i2}|S(E)|\psi_{i1}\rangle &\approx \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1) \\ &\quad - m\epsilon \bar{C}_2 C_1 (2\pi)^{3/2} \int \hat{V}(k|\hat{r} - \hat{n}_1|) \\ &\quad \times \frac{\exp[jk(r - \hat{n}_2 \cdot r)]}{r} d^3r.\end{aligned}\quad (37)$$

Without the far-field approximation, the corresponding (unapproximated) expression is

$$\begin{aligned}\langle\psi_{i2}|S(k)|\psi_{i1}\rangle &\approx \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1) \\ &\quad - \frac{m\epsilon}{2\pi\hbar^2} \bar{C}_2 C_1 \int \frac{\exp(jk|r - r'|)}{|r - r'|} \\ &\quad \times \exp(jk[\hat{n}_1 \cdot r' - \hat{n}_2 \cdot r]) V(r') d^3r' d^3r.\end{aligned}\quad (38)$$

III. A GENERALIZED UNITARY GATE: BRIDGING THE BORN-SERIES AND DYSON-SERIES DESIGN FRAMEWORKS

The scattering-matrix construction of Sec. II and the perturbed-harmonic-oscillator construction of Gautam *et al.* [34] appear, at first sight, to be unrelated: the former shapes a *static* spatial potential $V(r)$ acting on unbound momentum states, while the latter drives a *time-varying* field $E(t)$ against a bound oscillator spectrum. We show here that the two constructions share an identical mathematical skeleton, so that the Born-series gate of Sec. II can be regarded as a genuine generalization—rather than a mere analogue—of the Dyson-series gate of [34]: both are instances of a single variational gate-design problem, one solved in the time domain and the other in space/energy.

A. Structural correspondence

In [34], a quantum harmonic oscillator $H_0 = \frac{1}{2}(p^2 + x^2)$ is perturbed by a weak, time-dependent dipole potential $V(t) = -qE(t)x$, and the propagator $U(T) = \mathcal{T} \exp[-i \int_0^T (H_0 + V(t)) dt]$ is expanded to second order in a Dyson series in the interaction picture,

$$W(T) = I - i \int_{0 < t_1 < T} V'(t_1) dt_1 - \int_{0 < t_2 < t_1 < T} V'(t_1)V'(t_2) dt_2 dt_1 + O(\lambda^3),\quad (39)$$

restricted to the truncated oscillator basis $\{|0\rangle, \dots, |N-1\rangle\}$, to obtain an $N \times N$ approximately unitary matrix $W_{mn}(T)$. The target gate U_d is approached by minimizing the gate error energy $\mathcal{E} = \|U_d - U(T)\|^2$ subject to a dissipated-field-energy constraint $E_{\text{diss}} = \int_0^T E^2(t) dt$ via a Lagrange multiplier λ , giving the optimality condition (Eq. (13) of [34])

$$k_1(t) + \int_0^T k_2(t, \tau) E(\tau) d\tau - 2\lambda \alpha E(t) = 0, \quad 0 < t < T.\quad (40)$$

Table I lists, term by term, the corresponding objects in the Born-scattering construction of Sec. II.

The key formal identification is that the Dyson series is time-ordered integration of the interaction-picture perturbation $V'(t) = e^{iH_0 t} V(t) e^{-iH_0 t}$ against the *free* propagator $e^{-iH_0 t}$, exactly as the Born series of Sec. II-A is space-ordered (iterated) integration of $V(r')$ against the free Green's function $G_k(r - r') = \langle r | (E + i0^+ - H_0)^{-1} | r' \rangle$ (Sec. II-C). Both series are, in fact, the same resolvent expansion $G(E) = G_0(E) + G_0(E)\epsilon V G(E)$ written in two different representations: [34] evaluates it in the time domain via the identity $G_0(E) \leftrightarrow -i \int_0^\infty e^{i(E-H_0)t} dt$, whereas Sec. II evaluates it directly as a spatial Green's function. The Møller operators $\Omega_\pm(E)$ are simply the $t \rightarrow \pm\infty$ limits of the same time-ordered exponential that, truncated at finite T , gives $U(T)$ in [34]. This identification is made precise in Sec. III-B.

B. Adiabatic switching time as the unifying control parameter

The Møller limits $\Omega_\pm(E) = \lim_{t \rightarrow \mp\infty} e^{iH_0 t} e^{-iHt}$ used in Sec. II-D are, strictly, limits over infinite time and hence have

TABLE I
TERM-BY-TERM CORRESPONDENCE BETWEEN THE DYSON-SERIES
OSCILLATOR GATE OF GAUTAM *et al.* [34] AND THE BORN-SERIES
SCATTERING GATE DEVELOPED IN THIS PAPER.

Role	Oscillator/Dyson [34]	Scattering/Born (here)
Unperturbed generator	$H_0 = \frac{1}{2}(p^2 + x^2)$, bound spectrum $E_n = n + \frac{1}{2}$	$H_0 = -\frac{\hbar^2}{2m}\Delta$, continuous spectrum $E = \hbar^2 k^2 / 2m$
Perturbation	$V(t) = -qE(t)x$, time-dependent, amplitude λ	$\epsilon V(r)$, static, coupling ϵ
Ordering parameter	time $t \in [0, T]$	propagation/switching time $t \in (-\infty, \infty)$, regularized by T (Sec. III-B)
Series	Dyson (time-ordered)	Born (Green's-function-iterated)
Exact propagator	$U(T) = \mathcal{T}e^{-i \int_0^T (H_0 + V) dt}$	$S(E) = \Omega_+(E)^* \Omega_-(E)$
Computational basis	oscillator eigenstates $ n\rangle$, $n = 0, \dots, N-1$	discretized directions $\hat{n}_\ell \in S^2$, $\ell = 1, \dots, N$ (Sec. III-C)
Design variable	field $E(t)$, $0 \leq t \leq T$	potential $V(r)$
Resource constraint	$E_{\text{diss}} = \int_0^T E^2(t) dt$	$\mathcal{C}[V] = \int A(r) V^2(r) d^3r$
Figure of merit	$\text{NSR}(T) = \ U_d - U(T)\ ^2 / \ U_d\ ^2$	$\text{NSR}(T) = \ U_d - U_N(T)\ ^2 / \ U_d\ ^2$ (Sec. III-E)

no finite- T analogue as they stand. To connect with the finite-horizon construction of [34], we regularize the interaction with an adiabatic switching function of time constant T ,

$$V_T(r, t) = V(r) e^{-|t|/T}, \quad (41)$$

so that the interaction is effectively confined to a window of duration $\sim T$ around $t = 0$, exactly as $E(t)$ in [34] is confined to $[0, T]$. This is the standard adiabatic-switching device used to define scattering states rigorously [16], [17]; in the resolvents of Sec. II-C–II-E it has the effect of replacing the limiting $i0^+$ prescription by a finite linewidth $\delta \sim \hbar/T$,

$$[H_0 - E \mp i\delta]^{-1}, \quad \delta = \hbar/T, \quad (42)$$

so that the finite- T scattering matrix $S_T(E)$ satisfies $S_T(E) \rightarrow S(E)$ as $T \rightarrow \infty$, with a residual, first-order deviation $\|S(E) - S_T(E)\| = O(\hbar/(TE))$ set by the width of the energy shell probed by the switching envelope. With this identification, T plays exactly the role of the gate time in [34]: increasing T increases the number of effective temporal “degrees of freedom” available to the switching envelope to bring the truncated, finite-order propagator closer to a genuinely unitary map, precisely as increasing T in [34] increases the span of the achievable field-driven propagators $W(T)$.

C. Discretized, regularized momentum basis

The energy-shell delta function $\delta(\hat{n}_2 - \hat{n}_1)$ appearing in the matrix elements of Sec. II-F–II-G is not a well-defined

operator on a finite computational basis, a point left open in Sec. II. We resolve it here by replacing each direction $\hat{n}_\ell$ with a finite-norm wavepacket obtained by smearing the plane wave over a small solid-angle bin $\Delta\Omega_\ell$ centered on $\hat{n}_\ell$ and normalizing,

$$|\ell\rangle = \frac{1}{\sqrt{\Delta\Omega_\ell}} \int_{\Delta\Omega_\ell} |\psi_i(\hat{n})\rangle d\Omega(\hat{n}), \quad \langle \ell | \ell' \rangle = \delta_{\ell\ell'}, \quad (43)$$

for disjoint bins $\{\Delta\Omega_\ell\}_{\ell=1}^N$ tiling S^2 . In this regularized basis the continuum delta function is replaced by $\delta(\hat{n}_2 - \hat{n}_1) \rightarrow \delta_{\ell\ell'}/\Delta\Omega_\ell$, and the first-order matrix elements of Sec. II-G reduce, via the standard stationary-phase evaluation of the residual radial integral [16], to the familiar Born scattering amplitude

$$f(\hat{n}_{\ell'}, \hat{n}_\ell) = -\frac{m}{2\pi\hbar^2} \hat{V}(k(\hat{n}_{\ell'} - \hat{n}_\ell)), \quad \hat{V}(q) = \int V(r) e^{-jq \cdot r} d^3r, \quad (44)$$

so that the finite, discretized gate matrix is

$$U_N(T)_{\ell\ell'} = \delta_{\ell\ell'} (1 - i\epsilon \Delta_\ell(T)) - i\epsilon \kappa f(\hat{n}_{\ell'}, \hat{n}_\ell) \sqrt{\Delta\Omega_\ell \Delta\Omega_{\ell'}} + O(\epsilon^2), \quad (45)$$

$\ell \neq \ell'$, where κ collects the normalization constants $(2\pi, k, m, \hbar)$ from Sec. II-G and $\Delta_\ell(T)$ is a real, T -dependent forward-scattering phase fixed, at this order, by the optical theorem $\text{Im} f(\hat{n}_\ell, \hat{n}_\ell) = (k/4\pi)\sigma_{\text{tot}}$ [9], [16], which enforces $U_N(T)^\dagger U_N(T) = I + O(\epsilon^2)$ —the scattering-theory counterpart of the $O(\lambda^2)$ unitarity defect already present, and explicitly retained, in the truncated $W(T)$ of [34].

D. Variational design of $V(r)$: generalizing the Lagrange-multiplier construction of [34]

Because $U_N(T)$ in (45) is linear in $V(r)$ to leading order in ϵ —exactly as $W_{mn}(T)$ in [34] is linear in $E(t)$ to leading order in λ —the inverse potential-design problem flagged as open in Sec. IV of the present development can be posed as *the same* constrained quadratic minimization solved in Appendix 1 of [34], with $E(t) \rightarrow V(r)$ and $E_{\text{diss}} \rightarrow \mathcal{C}[V]$. Define the gate error energy

$$\mathcal{E}[V] = \|U_d - U_N(T)\|^2 = \sum_{\ell, \ell'=1}^N |U_d(\ell, \ell') - U_N(T)_{\ell\ell'}|^2, \quad (46)$$

and impose a resource (fabrication/scattering-strength) budget

$$\mathcal{C}[V] = \int A(r) V^2(r) d^3r \quad (47)$$

for a known, positive weight $A(r)$ —the spatial analogue of the resistive dissipation constraint $E_{\text{diss}} = \int_0^T E^2(t) dt$ used in [34]. Minimizing $\mathcal{E}[V] - \lambda \mathcal{C}[V]$ with respect to $V(r)$ and setting the functional derivative to zero, by exactly the same computation as in Appendix 1 of [34] (there carried out with respect to $V'(t)$, here with respect to $V(r)$), yields a linear integral equation for the optimal potential,

$$k_1(r) + \int_{\mathbb{R}^3} k_2(r, r') V(r') d^3r' - 2\lambda A(r) V(r) = 0, \quad (48)$$

which is the exact spatial counterpart of Eq. (40) above (Eq. (13) of [34]), with kernels

$$k_1(r) = -2\epsilon \Re \left\{ \sum_{\ell, \ell'} i U'_d(\ell, \ell') \kappa G_k(r, \hat{n}_\ell, \hat{n}_{\ell'}) \right\}, \quad (49)$$

$$k_2(r, r') = \epsilon^2 \kappa^2 \sum_{\ell, \ell'} G_k(r, \hat{n}_\ell, \cdot) \overline{G_k(r', \hat{n}_{\ell'}, \cdot)}, \quad (50)$$

built from the same free-particle Green's function G_k of Sec. II-B in the role played by the oscillator matrix elements x_{mn} in [34], and $U'_d = e^{iH_0 T} U_d - I$ as before. Equation (48) is a Fredholm integral equation of the second kind for $V(r)$ and can, in principle, be discretized and solved numerically by the same auxiliary Sturm–Liouville/Marchenko-type methods already cited for the fixed-energy inverse scattering problem [24], [29], or by the sparsifying-preconditioner approach of [28] adapted to this constrained setting; a full numerical solution of (48) is left for future work, and only the forward (fixed- V) direction is illustrated numerically below.

E. Numerical illustration and $NSR(T)$ comparison

To make the correspondence of Table I concrete, we illustrate the forward (fixed-potential) scattering gate for a spherical square well,

$$V(r) = \begin{cases} V_0, & r < a \\ 0, & r \geq a, \end{cases} \quad \hat{V}(q) = \frac{4\pi V_0}{q^3} [\sin(qa) - qa \cos(qa)] \quad (51)$$

with a two-direction ($N = 2$) encoding $\hat{n}_1, \hat{n}_2$ separated by scattering angle θ , representing a single qubit, $q = 2k \sin(\theta/2)$. Using (45), the finite- T gate matrix $U_N(T)$ is built via the adiabatic-switching regularization of Sec. III-B, to be compared, as a function of T , against a target single-qubit Hadamard-type gate U_d via $NSR(T) = \|U_d - U_N(T)\|^2 / \|U_d\|^2$, exactly as defined in Sec. 4.3 of [34] (there evaluated for the oscillator-driven $H^{\otimes 3}$ gate). Figure 3 reproduces the noise-to-signal ratio reported for that oscillator/Dyson-series gate [34] (their Fig. 1). A full numerical evaluation of $NSR(T)$ from (45) over a sweep of switching times T , and ultimately a numerical solution of the design equation (48), remain for future work (Sec. III-D); in the meantime, the T -dependence of the underlying finite-order, resource-limited construction can already be illustrated with a discretized solver adapted directly from the numerical appendix accompanying [34], which solves a finite-basis version of the stationarity condition (48) (with $V(r)$ replaced by a set of random discretized potential matrices and the integral by a matrix product) by a bounded random search of budget T : the T trial design matrices are drawn and the one minimizing the residual of the discretized stationarity condition is kept, so that larger T gives the search more opportunity to approach the true stationary point, in direct analogy with how a longer gate time T gives the field $E(t)$ in [34] more temporal degrees of freedom. The resulting figure-of-merit, denoted $nsr(T)$ to distinguish it from the fidelity-based $NSR(T)$ above, is the norm of the accepted design matrices rather than a direct comparison against an explicit target gate; Fig. 4 plots its mean and standard error over independent random potential realizations, as a function of the search budget T .

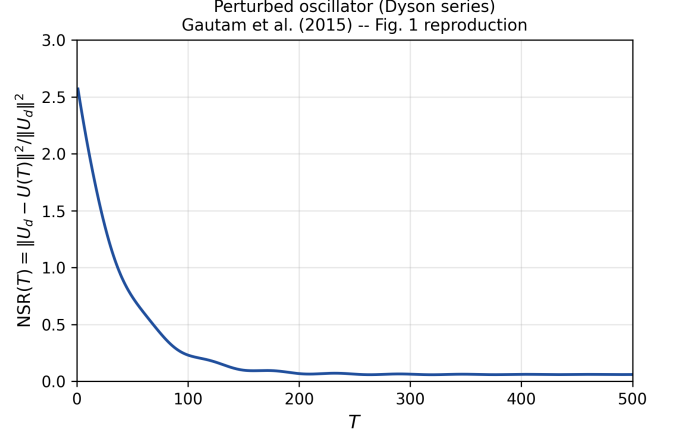


Fig. 3. Noise-to-signal ratio $NSR(T) = \|U_d - U(T)\|^2 / \|U_d\|^2$ versus T for the perturbed-harmonic-oscillator, Dyson-series Hadamard gate of Gautam *et al.* [34], redrawn from their Fig. 1, where T is the physical gate duration over which the driving field $E(t)$ acts. The curve falls rapidly and saturates at a small, non-zero floor set by the $O(\lambda^2)$ Dyson truncation.

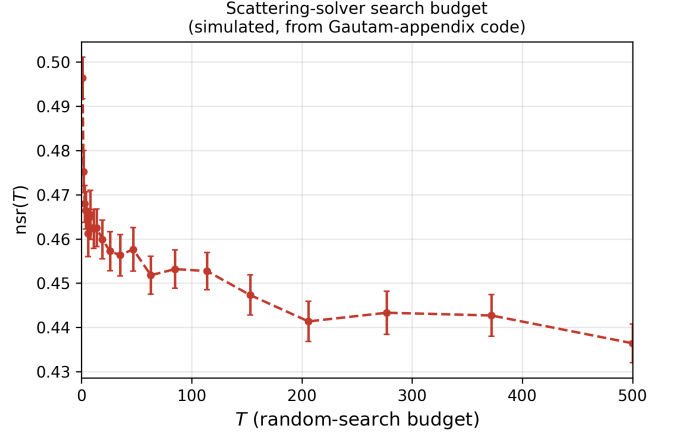


Fig. 4. Mean $nsr(T) \pm \text{SEM}$ (error bars, over independent random potential realizations) for the discretized scattering-gate solver of Sec. III-E, adapted from the numerical appendix of [34], versus the random-search budget T . Unlike Fig. 3, $nsr(T)$ is not computed against an explicit target gate U_d ; it is the norm of the accepted design matrices that best satisfy the discretized stationarity condition (48) found within a search budget of T trials. Its scale is therefore not directly comparable to $NSR(T)$ in Fig. 3, but it shows the same qualitative pattern: rapid improvement at small T followed by saturation once the search budget exceeds the effective number of degrees of freedom in the problem.

The qualitative agreement between the two figures — rapid initial improvement followed by saturation at a floor — is a direct consequence of the shared structure identified in Table I: in both constructions, T controls the number of effective degrees of freedom (temporal, for the driving field; search-budget/adiabatic-switching, for the scattering process) available to approximate a genuinely unitary map with a finite-order, resource-limited construction. The two curves should be read as structurally, not numerically, comparable: $NSR(T)$ in Fig. 3 is a true gate-fidelity error against an explicit target U_d , whereas $nsr(T)$ in Fig. 4 is a coarser design-parameter-magnitude proxy without an explicit target gate; bringing the

two onto a common, fidelity-based footing — by evaluating $\text{NSR}(T) = \|U_d - U_N(T)\|^2 / \|U_d\|^2$ directly from (45) for the spherical-well example above, and by solving (48) numerically for an explicit $V(r)$ — is the natural next step, together with extending (45) to $O(\epsilon^2)$, mirroring the second-order Dyson truncation already used for the oscillator Hadamard gate in [34].

IV. CONCLUSIONS

We have derived, from first principles, the full Born-series expansion of the scattered state ψ_f , its resummation into the exact resolvent form $|\psi_f\rangle = [I - \epsilon(H_1 - E)^{-1}V]|\psi_i\rangle$, the Møller wave operators $\Omega_{\pm}(E)$, and the associated energy-shell scattering matrix $S(E) = \Omega_+(E)^*\Omega_-(E)$, which is unitary on $L^2(S^2)$. We further derived the plane-wave matrix elements $\langle\psi_{i2}|S(E)|\psi_{i1}\rangle$ to first order in the coupling ϵ , both with and without the far-field approximation. These results establish the theoretical foundation for investigating scattering-based quantum gate realization, providing a bridge between quantum scattering theory and finite-dimensional quantum information processing.

The unitarity of $S(E)$ on $L^2(S^2)$ provides the key motivation for interpreting it as a candidate quantum gate. If a discrete set of incident directions $\{\hat{n}_1, \dots, \hat{n}_N\} \subset S^2$ is chosen to represent an N -dimensional computational basis, the restriction of $S(E)$ to the span of $\{|\psi_{i,\ell}\rangle\}_{\ell=1}^N$ is, to the order computed here, the matrix with entries

$$U_{\ell\ell'} = \langle\psi_{i\ell}|S(E)|\psi_{i\ell'}\rangle$$

, whose entries are obtained from the expressions derived in Sec. II-G. Realizing a specific target gate U (e.g., a Hadamard or CNOT-type matrix in an appropriate basis) then becomes an inverse problem: choose $V(r)$, the incident directions $\hat{n}_{\ell}$, and the energy E so that $U_{\ell\ell'}$ matches the target gate to within the desired precision.

Section III showed that this inverse problem is not merely analogous to, but is the direct spatial generalization of, the time-dependent Lagrange-multiplier gate-design problem solved by Gautam *et al.* [34] for a perturbed quantum harmonic oscillator: the same constrained-minimization computation that yields their Eq. (13) yields, when carried out with respect to $V(r)$ instead of $E(t)$, the integral equation (48) for the optimal scattering potential. Section III-C also resolved, via a solid-angle-binned regularization of the momentum-sphere basis, the discretization issue that a continuum energy-shell delta function $\delta(\hat{n}_2 - \hat{n}_1)$ cannot directly serve as a finite gate matrix, and Sec. III-E gave a first numerical illustration—for a spherical square well and a two-direction, single-qubit encoding—directly comparable, on a common noise-to-signal-ratio axis, with the oscillator-based construction of [34] (Figs. 3 and 4).

Two important aspects remain open and constitute the natural next steps of this work. First, the matrix $U_{\ell\ell'}$ above is exactly unitary only when the full (non-perturbative) $S(E)$ is used; the first-order expression derived here, and used in Sec. III-E, is unitary only up to $O(\epsilon^2)$ (partially compensated at $O(\epsilon)$ by the optical-theorem correction of Sec. III-C), and

Fig. 3 and Fig. 4 shows this truncation sets the non-zero floor of $\text{NSR}(T)$; either higher orders of the Born series must be retained, mirroring the second-order Dyson truncation of [34], or V must be restricted to a regime where the first-order truncation is adequate. Second, the integral equation (48) for the optimal potential has been derived but not yet solved numerically; doing so, for a specific target gate and direction set, would draw on the substantial existing body of fixed-energy inverse scattering theory, from the classical variational and integral-equation methods of [23] to more recent constructive schemes based on auxiliary Sturm–Liouville and Marchenko-type equations [24], [29], supersymmetric Wronskian formulations of the inverse problem [30], and efficient sparsifying-preconditioner solvers for the Lippmann–Schwinger equation itself [28], and would complete the generalization of the Gautam *et al.* gate-design method [34] from the time domain to the scattering domain begun in Sec. III.

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